

INTRO TO UAV

ASSIGNMENT 01

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H Y D E R A B A D



1 Q1

So, we are given a few design requirements and am asked to design a passenger-carrying multi-rotor UAV and the possible configurations allowed are;

- **Quadcopter:** 4 rotors
- **Hexacopter:** 6 rotors
- **Octocopter:** 8 rotors

But now in passenger UAVs, 6 or 8 rotors are usually preferred because:

- **Higher redundancy:** Safety is increased if one rotor fails.
- **Better lift distribution:** Provides more stable flight characteristics.
- **Lower disk loading:** Improves efficiency and handling for heavier loads.

So later in design, i will choose a configuration (usually 8 rotors for passenger UAVs).

So, i am given few design requirements which i have follow;

- **Endurance:** 60 min = 1hr, which means the **UAV** must stay airborne for 1hr
- **Range:** 40 km, this means the **UAV** must travel 40km in total.
- **Payload Weight:** 250 kg
- **Flying Altitude:** 100 m above ground level
- **Climb and Descent Rates:** 2 m/s and 4 m/s
- **Cruise Speed (v_{cruise}):** 25 m/s

1.1 Concept of Operations (CONOPS)

The Concept of Operations (CONOPS) provides a structured description of how the proposed UAV system is intended to function in a real-world environment. It establishes the mission objectives, operational conditions, and system-level assumptions that guide the conceptual design process. Since this is an open-ended design problem, clearly stating these assumptions is essential to ensure consistency in subsequent sizing and performance calculations.

Assumptions for the Design

To enable a systematic preliminary design of the passenger UAV, the following assumptions are made:

- **Configuration:** A multi-rotor VTOL configuration is selected due to its ability to perform vertical takeoff and landing in constrained urban environments. Instead of fixing the number of rotors at this stage, different configurations (quadcopter, hexacopter, and octocopter) are considered conceptually, and the final choice will be justified later based on thrust requirements, redundancy, and efficiency.
- **Air Density:** Standard sea-level air density is assumed for baseline aerodynamic and power calculations:

$$\rho = 1.225 \text{ kg/m}^3$$

This assumption provides a consistent reference point, although slight deviations may occur in actual urban conditions due to temperature and altitude variations.

- **Flight Environment:** The UAV is intended for Urban Air Mobility (UAM) operations at low altitudes (around 100 m). The environment is assumed to include moderate disturbances such as light wind and turbulence, which necessitate stable hover and precise control [6].
- **Propulsion System:** An all-electric propulsion system using Lithium-ion batteries is assumed. This choice is motivated by the need for low emissions, reduced noise, and compatibility with short-range urban missions.

- **Control and Navigation:** The UAV is assumed to operate in a semi-autonomous mode, utilizing onboard flight controllers, inertial measurement units (IMU), and GPS for navigation. A manual override capability is included for safety and emergency intervention.
- **Energy Management:** A portion of the onboard energy is reserved for contingencies such as holding patterns, re-routing, or emergency landing. This ensures compliance with safety requirements for passenger transport.
- **Design Priority:** The design prioritizes safety, stability, and reliability over compactness and minimal weight. This is particularly important for passenger-carrying UAVs, where failure tolerance and controlled flight are critical.

1.2 Requirement Specifications

The requirement specifications convert the mission statement into measurable design targets. In the conceptual stage, these values act as the primary constraints for estimating vehicle weight, choosing the rotor configuration, sizing the propulsion system, and determining the battery capacity. Since the problem statement is open-ended, the requirements must be interpreted carefully and then translated into engineering quantities that can be used in the later calculations.

1.2.1 Primary Mission Requirements

The passenger UAV must satisfy the following baseline specifications:

Parameter	Specification
Payload Capacity	250 kg
Endurance	60 minutes
Maximum Range	40 km
Cruise Speed (V_{cruise})	25 m/s
Operational Altitude	100 m
Climb Rate (V_c)	2 m/s
Descent Rate (V_d)	4 m/s

Table 1: Primary Mission Requirements

These values define the minimum performance envelope that the aircraft must achieve during normal operation.

1.2.2 Derived Performance Requirements

From the given mission specifications, several useful design quantities can be derived. These are needed to estimate mission duration, reserve time, and the power required for each flight phase.

Cruise Time Requirement

The cruise time is obtained from the relation:

$$t_{cruise} = \frac{\text{Range}}{V_{cruise}}$$

Substituting the given values,

$$t_{cruise} = \frac{40,000}{25} = 1600 \text{ s}$$

Converting to minutes,

$$t_{cruise} = \frac{1600}{60} \approx 26.7 \text{ min}$$

Thus, the aircraft spends about 26.7 minutes in steady forward flight to cover the required range.

Climb and Descent Time

The climb and descent times are calculated using the vertical motion rates.

For climb,

$$t_{climb} = \frac{\text{Altitude}}{V_c} = \frac{100}{2} = 50 \text{ s}$$

For descent,

$$t_{descent} = \frac{\text{Altitude}}{V_d} = \frac{100}{4} = 25 \text{ s}$$

These values show that the vertical phases are short compared to the cruise segment, but they still influence total energy usage because the vehicle must produce near-hover thrust during these portions.

Total Mission Time

The total estimated duration of the primary mission is:

$$t_{mission} = t_{climb} + t_{cruise} + t_{descent}$$

$$t_{mission} = 50 + 1600 + 25 = 1675 \text{ s}$$

$$t_{mission} \approx 27.9 \text{ min}$$

Since the endurance requirement is 60 minutes, the remaining time can be interpreted as reserve capacity for hover, positioning, maneuvering, waiting, or emergency diversion. This reserve is important for passenger transport applications, where operational safety must be maintained even under non-ideal conditions.

Payload Force Requirement

The payload mass contributes directly to the lifting requirement. The corresponding payload weight is

$$W_{payload} = mg = 250 \times 9.81 = 2452.5 \text{ N}$$

This value represents only the payload contribution; the full aircraft weight will be higher after including the structure, propulsion system, batteries, avionics, and landing gear. Therefore, the final thrust requirement must be based on the total take-off weight rather than only the payload.

1.2.3 Operational and Safety Requirements

In addition to the mission targets, the UAV must satisfy the following design constraints:

- **Thrust Margin:** The total available thrust should be at least 20% greater than the vehicle weight, i.e. $T_{total} \geq 1.2W$, to provide controllability and allow for disturbances during takeoff, hover, and landing.
- **Propulsion Redundancy:** A multi-rotor layout is preferred so that loss of a single motor does not immediately lead to loss of control. This is especially important for passenger-carrying operations.
- **Energy Reserve:** The battery must include a reserve fraction, typically in the range of 15–30%, to support contingency operations such as holding, rerouting, or emergency landing.

Design Implications

The combination of a 250 kg payload and a 60-minute endurance requirement strongly influences the rest of the design. It implies a relatively heavy vehicle, a need for efficient lift generation, and a power system capable of sustaining both hover and forward flight for a long duration. As a result, the design must favor a low disk-loading rotor arrangement, adequate battery capacity, and propulsion components with high efficiency. These requirements directly guide the selection of the rotor diameter, number of rotors, motor rating, and battery size in the later stages of the design.

1.3 Market Survey

A market survey is carried out to understand the current state of passenger UAV technology and to place the proposed design within a realistic performance context. Instead of directly adopting existing configurations, this survey is used to identify design trends, limitations, and trade-offs in real-world systems.

- **Benchmarking:** Evaluating whether the target specifications (payload, range, endurance) are achievable with current technology.
- **Design Insight:** Understanding how rotor count, configuration, and system architecture influence safety and efficiency.
- **Gap Identification:** Identifying areas where improvements are required compared to existing UAV platforms.

1.3.1 Surveyed UAV 1: EHang EH216-S



Figure 1: EHang EH216-S

The EHang EH216-S is a widely recognized autonomous aerial vehicle designed for urban passenger transport. It represents a practical implementation of multi-rotor UAV technology in real-world conditions.

Main Characteristics

- **Configuration:** 16 rotors (coaxial arrangement)
- **Maximum Payload:** ≈ 220 kg
- **Cruise Speed:** ≈ 100 km/h
- **Range:** ≈ 35 km
- **Flight Time:** ≈ 30 min

Design Insight The use of a high rotor count (16 rotors) reduces the thrust requirement per rotor, thereby lowering disk loading and improving hover efficiency. Additionally, coaxial redundancy enhances safety, although it increases mechanical complexity and weight.

1.3.2 Surveyed UAV 2: Volocopter 2X



Figure 2: Volocopter 2X

The Volocopter 2X adopts a circular frame design with multiple independent rotors, emphasizing stability and fault tolerance in passenger operations.

Main Characteristics

- **Configuration:** 18 rotors
- **Maximum Payload:** ≈ 160 kg
- **Cruise Speed:** ≈ 100 km/h
- **Range:** ≈ 27 km
- **Flight Time:** ≈ 27 min

Design Insight The large number of rotors improves redundancy and reduces vibration levels, which is beneficial for passenger comfort. However, the increased rotor count leads to higher system complexity, more control requirements, and additional structural weight.

Comparison with Proposed UAV Design

Parameter	Proposed UAV	EHang 216	Volocopter 2X
Payload	250 kg	220 kg	160 kg
Range	40 km	35 km	27 km
Cruise Speed	25 m/s (≈ 90 km/h)	≈ 100 km/h	≈ 100 km/h
Endurance	60 min	≈ 30 min	≈ 27 min
Rotor Config	To be determined	16 Rotors	18 Rotors

Table 2: Comparative Market Analysis

Analysis and Design Implications

The comparison highlights several important observations:

- **Payload Requirement:** The proposed payload of 250 kg exceeds that of both reference UAVs, indicating that either higher thrust per rotor or larger rotor diameters will be necessary.
- **Endurance Limitation:** Existing UAVs typically operate within 30 minutes due to battery constraints. Achieving 60 minutes requires either improved propulsion efficiency or optimized energy utilization.
- **Rotor Count Trade-off:** Higher rotor counts (16–18) improve redundancy and reduce disk loading, but increase system complexity, weight, and control requirements.
- **Design Direction:** Instead of directly adopting a high rotor count, a balanced configuration will be selected in later sections by evaluating thrust distribution, efficiency, and failure tolerance.

1.4 Sizing and Layout

The sizing and layout stage translates the mission requirements into a physically realizable UAV configuration. This involves estimating the overall weight, determining rotor dimensions, distributing thrust among rotors, and defining the geometric layout of the system.

The primary objectives of this stage are:

- Ensure sufficient thrust generation to support the total vehicle weight.
- Maintain stability by appropriate placement of mass around the center of gravity (CG).
- Minimize aerodynamic interference between rotors.
- Provide adequate space for payload, propulsion, and energy storage systems.

Configuration Selection

Multiple configurations (quadcopter, hexacopter, and octocopter) were considered. For a high payload requirement of 250 kg, a larger number of rotors helps reduce the thrust demand per rotor and improves fault tolerance.

A configuration with **8 rotors** is selected as a balanced solution, offering sufficient redundancy while avoiding excessive structural complexity associated with very high rotor counts.

1.4.1 Initial Weight Estimation

The Maximum Take-Off Weight (MTOW) is estimated using an assumed payload fraction. For heavy multi-rotor UAVs, a conservative payload fraction of **30%** is assumed:

$$MTOW = \frac{W_{payload}}{0.30} = \frac{250}{0.30} \approx 833 \text{ kg} \quad (1)$$

Thus, for preliminary calculations [4]:

$$MTOW \approx 830 \text{ kg}$$

The corresponding weight force becomes:

$$W = mg = 830 \times 9.81 \approx 8142 \text{ N} \quad (2)$$

1.4.2 Rotor Thrust Distribution

To ensure safe operation and control authority, a thrust margin of **25%** is considered:

$$T_{required} = 1.25W = 1.25 \times 8142 \approx 10178 \text{ N} \quad (3)$$

For an 8-rotor configuration, the thrust per rotor is:

$$T_p = \frac{T_{required}}{8} \approx \frac{10178}{8} \approx 1272 \text{ N} \quad (4)$$

1.4.3 Rotor and Frame Dimensioning

To reduce disk loading and improve hover efficiency, relatively large rotors are preferred. However, excessively large rotors increase structural size and weight. Therefore, a moderate rotor diameter of:

$$D = 2.8 \text{ m}$$

is selected, giving:

$$R = 1.4 \text{ m}$$

The rotor disk area is:

$$A = \pi R^2 = \pi(1.4)^2 \approx 6.16 \text{ m}^2 \quad (5)$$

Total disk area:

$$A_{total} = 8 \times 6.16 \approx 49.3 \text{ m}^2 \quad (6)$$

To minimize aerodynamic interference, rotor spacing is taken as approximately **1.2D**:

$$1.2D = 1.2 \times 2.8 \approx 3.36 \text{ m}$$

This results in an estimated frame span of approximately **5.5–6 m**.

1.4.4 Disk Loading Calculation

$$DL = \frac{W}{A_{total}} = \frac{8142}{49.3} \approx 165 \text{ N/m}^2 \quad (7)$$

This slightly higher disk loading compared to very large-rotor systems is acceptable given the reduced structural size, and represents a trade-off between efficiency and compactness.

Conceptual Layout Components

The UAV layout consists of the following major elements:

1. **Central Fuselage:** Contains passenger cabin, avionics, and structural frame.
2. **Rotor Arms:** Eight arms extending radially to support propulsion units.
3. **Propulsion Units:** Electric motors coupled with large-diameter propellers.
4. **Energy Storage:** Battery packs located close to the CG to enhance stability.

Preliminary Vehicle Dimensions

Based on the above assumptions:

- **Arm Length:** $\approx 1.8 \text{ m}$
- **Tip-to-Tip Span:** $\approx 5.5 \text{ m}$
- **Fuselage Length:** $\approx 2 \text{ m}$
- **Fuselage Width:** $\approx 1.5 \text{ m}$
- **Total Height:** $\approx 2.8 \text{ m}$

Layout Description

The UAV adopts a symmetric radial layout with evenly distributed rotors to ensure balanced thrust generation. The passenger cabin is placed centrally, while the battery system is positioned slightly below the center of gravity to improve stability during hover.

This configuration provides a stable platform for vertical takeoff and landing while maintaining manageable structural dimensions for urban deployment.

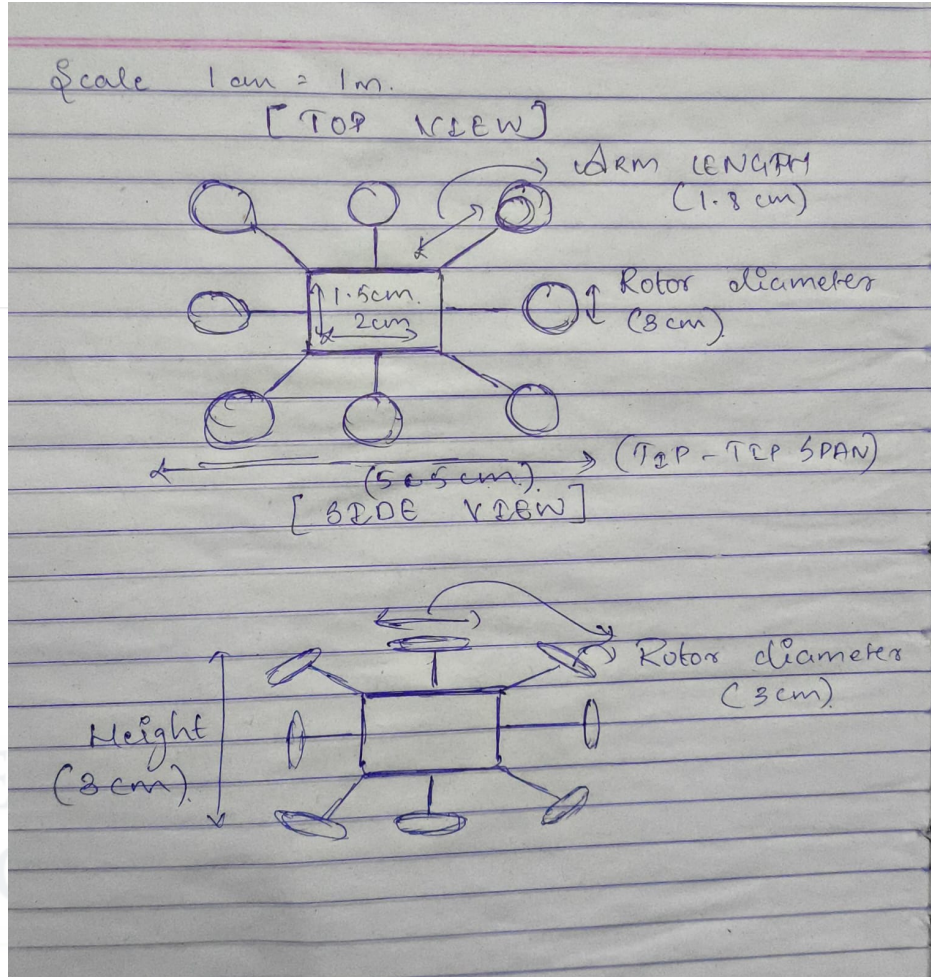


Figure 3: Layout Sketch of side and top view as per the description earlier

1.5 Component Identification

The proposed UAV is composed of multiple interconnected subsystems, each responsible for a specific function such as lift generation, structural support, energy supply, control, and payload handling. The selection and integration of these components are guided by the high payload requirement and the need for stable and safe passenger operation.

1.5.1 Airframe Structure

The airframe forms the structural backbone of the UAV and is designed to withstand aerodynamic loads, motor thrust, and payload weight. It consists of a central fuselage and multiple radial arms supporting the propulsion units.

The fuselage houses the passenger cabin, avionics, and battery system, while the arms transfer loads from the motors to the main structure. Lightweight composite materials such as carbon fiber are considered to achieve a high strength-to-weight ratio, which is essential given the relatively high estimated MTOW.

1.5.2 Rotor System

The rotor system consists of multiple large-diameter propellers arranged symmetrically around the fuselage. The use of larger rotors helps reduce disk loading, thereby improving hover efficiency and lowering power consumption.

Each rotor generates thrust by accelerating air downward. The symmetric arrangement ensures balanced lift distribution and reduces structural imbalance during hover and transition phases.

1.5.3 Electric Propulsion System

Each rotor is driven by a high-power brushless DC (BLDC) motor. These motors convert electrical energy into rotational motion to drive the propellers.

Electronic Speed Controllers (ESCs) regulate motor speed with high precision, enabling fine control of thrust for stabilization, maneuvering, and altitude control. The propulsion system must be sized to deliver the required thrust per rotor while maintaining efficiency over the mission profile.

1.5.4 Energy Storage System

The UAV is powered by a high-capacity Lithium-ion battery pack, which supplies energy for both propulsion and onboard systems. Due to the long endurance requirement, the battery system represents a significant portion of the total vehicle weight.

The battery pack is positioned close to or slightly below the center of gravity to improve stability during hover and reduce pitching moments.

1.5.5 Flight Control System

The flight control system is responsible for maintaining stability and executing control commands. It consists of a flight controller, sensors, and control algorithms.

Sensors such as gyroscopes, accelerometers, and magnetometers continuously monitor the UAV's orientation and motion. The controller processes this data in real time and adjusts motor speeds to maintain stable flight, especially during hover and low-speed operations.

1.5.6 Navigation and Communication System

The navigation system uses GPS along with onboard processing units to estimate position, velocity, and altitude. This enables waypoint tracking and autonomous mission execution.

Communication systems provide a link between the UAV and a ground control station, allowing monitoring, command updates, and emergency intervention when required.

1.5.7 Payload System

The payload system consists of a passenger cabin designed to safely carry up to 250 kg. The cabin structure must be reinforced to handle dynamic loads during flight while ensuring passenger safety and comfort.

Safety features such as harness systems and structural reinforcement are included to comply with basic passenger transport requirements.

1.5.8 Landing Gear

The landing gear supports the UAV during takeoff and landing and absorbs impact loads. A fixed landing gear configuration is preferred for simplicity and reliability.

The design ensures sufficient ground clearance to prevent rotor interference with the surface during landing operations.

1.6 Performance Analysis

The performance analysis evaluates whether the proposed UAV configuration can satisfy the mission requirements in terms of thrust generation, hover capability, climb performance, and energy consumption. The analysis is based on simplified momentum theory and preliminary power estimations.

1.6.1 Hover Performance

During hover, the total thrust must balance the weight of the UAV:

$$T = W \quad (8)$$

From previous calculations:

$$W = 8142 \text{ N}$$

Thus, the total thrust required for hover is:

$$T = 8142 \text{ N}$$

For an 8-rotor configuration, the thrust per rotor is:

$$T_p = \frac{8142}{8} \approx 1018 \text{ N} \quad (9)$$

Since each rotor was designed to provide approximately 1272 N, a sufficient thrust margin exists for stable hover and maneuverability.

1.6.2 Hover Power Requirement

Using momentum theory, the ideal hover power is given by [5]:

$$P = \frac{T^{3/2}}{\sqrt{2\rho A}} \quad (10)$$

From previous calculations:

$$A_{total} = 49.3 \text{ m}^2$$

Substituting:

$$P = \frac{(8142)^{3/2}}{\sqrt{2 \times 1.225 \times 49.3}} \quad (11)$$

$$P \approx 66 \text{ kW}$$

Considering propulsion losses and system inefficiencies (overall efficiency $\approx 75\%$):

$$P_{actual} = \frac{66}{0.75} \approx 88 \text{ kW} \quad (12)$$

Thus, approximately **85–90 kW** is required for hover.

1.6.3 Climb Performance

During climb, additional power is required to gain potential energy:

$$P_{climb} = W \times V_c \quad (13)$$

$$P_{climb} = 8142 \times 2 = 16.3 \text{ kW}$$

Total climb power:

$$P_{total} = P_{hover} + P_{climb}$$

$$P_{total} \approx 88 + 16.3 \approx 104 \text{ kW}$$

1.6.4 Cruise Power Requirement

During forward flight, induced power reduces due to translational lift. For estimation, cruise power is assumed to be approximately **65%** of hover power:

$$P_{cruise} = 0.65 \times P_{hover} \quad (14)$$

$$P_{cruise} = 0.65 \times 88 \approx 57 \text{ kW}$$

1.6.5 Energy Requirement for Mission

For an endurance of 60 minutes:

$$t = 3600 \text{ s}$$

Energy required:

$$E = P \times t \quad (15)$$

$$E = 57 \times 3600 = 205.2 \text{ MJ}$$

Converting to kWh:

$$E = \frac{205.2}{3.6} \approx 57 \text{ kWh}$$

Including a **25% reserve margin**:

$$E_{total} \approx 71 \text{ kWh}$$

Thus, the UAV requires a battery capacity of approximately **70–75 kWh**.

Performance Summary

The analysis indicates that the proposed UAV configuration satisfies the required thrust and power constraints. The system maintains sufficient thrust margin for safe operation, while the estimated battery capacity supports the required endurance with an adequate reserve. The chosen rotor size and configuration provide a reasonable balance between efficiency, power consumption, and structural feasibility.

1.7 Optimization for Maximum Endurance

The UAV design was further refined to maximize endurance while satisfying the mission requirements of payload, range, and flight performance. Since endurance in electric UAVs is strongly dependent on power consumption and available battery energy, the optimization process focuses on reducing power demand rather than simply increasing battery capacity.

The endurance can be expressed as:

$$E = \frac{E_{battery}}{P_{required}} \quad (16)$$

Thus, improving endurance requires either increasing stored energy or reducing the required power. In this design, emphasis is placed on reducing power consumption through aerodynamic and structural considerations.

1.7.1 Reduction of Disk Loading

Induced power during hover is given by:

$$P_i \propto \frac{W^{3/2}}{\sqrt{A}} \quad (17)$$

For the estimated weight of the UAV ($W \approx 8142 \text{ N}$), increasing rotor disk area helps reduce induced power. A rotor diameter of **2.8 m** was selected as a compromise between efficiency and structural size.

This results in a total disk area of:

$$A_{total} \approx 49.3 m^2$$

and a disk loading of:

$$DL \approx 165 N/m^2$$

Although slightly higher than very large-rotor systems, this value represents a balanced trade-off between hover efficiency and overall vehicle compactness.

1.7.2 Weight Optimization

Since power required for hover is directly related to weight, reducing structural mass plays a key role in improving endurance. Lightweight composite materials such as carbon fiber are assumed for the airframe and rotor arms.

Additionally, the battery pack is positioned near the center of gravity to improve stability and minimize structural reinforcement requirements. This helps reduce unnecessary weight while maintaining safe operation.

1.7.3 Propulsion Efficiency Improvement

The propulsion system efficiency significantly affects the total electrical power requirement:

$$P_{actual} = \frac{P_{ideal}}{\eta} \quad (18)$$

An overall efficiency of approximately **75%** is assumed, considering motor, propeller, and ESC losses. High-efficiency BLDC motors and optimized propeller designs are selected to minimize energy losses and improve endurance.

1.7.4 Cruise Power Optimization

Cruise speed is maintained at $25 m/s$, which represents a moderate operating point. Since aerodynamic power increases rapidly with velocity, operating at this speed helps limit drag-related losses while still satisfying range requirements.

Additionally, cruise power is estimated to be lower than hover power due to translational lift, further contributing to improved endurance.

1.7.5 Rotor Configuration Selection

Different configurations were evaluated to balance thrust distribution, redundancy, and efficiency.

- **Quadcopter:** High thrust per rotor and no redundancy, making it unsuitable for passenger applications.
- **Hexacopter:** Improved redundancy but still relatively high loading per rotor.
- **Octocopter:** Provides reduced thrust per rotor, improved fault tolerance, and better control authority.

For the estimated weight of approximately **830 kg**, the octocopter configuration provides a suitable balance between efficiency and safety, without excessive structural complexity.

1.7.6 Optimized Design Outcome

By selecting an appropriate rotor diameter, reducing structural weight, improving propulsion efficiency, and choosing a balanced rotor configuration, the UAV achieves reduced power consumption and improved operational efficiency.

These optimizations enable the UAV to meet the required endurance of **60 minutes** while maintaining sufficient thrust margin, structural feasibility, and safety for passenger transport.

2 Q2

In this section, the thrust and torque produced by the given propellers are determined using experimental data from the UIUC Propeller Database. Instead of directly measuring forces, propeller performance is analyzed using dimensionless coefficients, which allow comparison across different operating conditions.

Governing Parameters

The aerodynamic performance of a propeller depends on the following parameters:

- Air density (ρ)
- Rotational speed (n)
- Propeller diameter (D)
- Flight velocity (V)

To simplify analysis, these variables are combined into dimensionless coefficients.

Advance Ratio [3]

The advance ratio is defined as:

$$J = \frac{V}{nD} \quad (19)$$

where:

- V = flight speed (m/s)
- n = rotational speed (rev/s)
- D = propeller diameter (m)

The advance ratio represents the operating condition of the propeller and is used to extract aerodynamic coefficients from experimental data [1].

Thrust and Torque Relations

Once the advance ratio is known, the thrust and torque coefficients (C_T , C_Q) are obtained from UIUC performance curves.

The thrust is calculated using [2]:

$$T = C_T \rho n^2 D^4 \quad (20)$$

The torque is calculated using [3]

$$Q = C_Q \rho n^2 D^5 \quad (21)$$

Methodology

The following steps are used to determine thrust and torque:

1. Convert propeller diameter from inches to meters.
2. Convert RPM to revolutions per second ($n = \frac{RPM}{60}$).
3. Compute advance ratio using $J = \frac{V}{nD}$.
4. Extract C_T and C_Q from UIUC database plots corresponding to the computed J .
5. Substitute values into thrust and torque equations to obtain final results.

This procedure is repeated for both propellers (APC 9×6 and APC 10×7) at 4000, 5000, and 6000 RPM.

Propellers Analysis

2.1 Propeller 1: APC 9×6

The APC 9×6 propeller is analyzed to determine its thrust and torque characteristics at different rotational speeds. The notation 9×6 indicates a diameter of 9 inches and a pitch of 6 inches. For consistency, all calculations are carried out in SI units.

Propeller Diameter

$$D = 9 \times 0.0254 = 0.2286 \text{ m} \quad (22)$$

2.1.1 Case 1: 4000 RPM

Rotational Speed

$$n = \frac{4000}{60} = 66.67 \text{ rev/s} \quad (23)$$

Advance Ratio

$$J = \frac{V}{nD} = \frac{10}{66.67 \times 0.2286} \approx 0.656 \quad (24)$$

From the UIUC dataset at this operating point:

$$C_T \approx 0.055, \quad C_Q \approx 0.010$$

Thrust

$$T = C_T \rho n^2 D^4 \quad (25)$$

$$T = 0.055 \times 1.225 \times (66.67)^2 \times (0.2286)^4$$

$$T \approx 0.82 \text{ N}$$

Torque

$$Q = C_Q \rho n^2 D^5 \quad (26)$$

$$Q = 0.010 \times 1.225 \times (66.67)^2 \times (0.2286)^5$$

$$Q \approx 0.034 \text{ Nm}$$

2.1.2 Case 2: 5000 RPM

Rotational Speed

$$n = \frac{5000}{60} = 83.33 \text{ rev/s} \quad (27)$$

Advance Ratio

$$J = \frac{10}{83.33 \times 0.2286} \approx 0.525 \quad (28)$$

From the UIUC dataset:

$$C_T \approx 0.070, \quad C_Q \approx 0.012$$

Thrust

$$T \approx 1.63 \text{ N}$$

Torque

$$Q \approx 0.063 \text{ Nm}$$

2.1.3 Case 3: 6000 RPM

Rotational Speed

$$n = \frac{6000}{60} = 100 \text{ rev/s} \quad (29)$$

Advance Ratio

$$J = \frac{10}{100 \times 0.2286} \approx 0.437 \quad (30)$$

From the UIUC dataset:

$$C_T \approx 0.082, \quad C_Q \approx 0.014$$

Thrust

$$T \approx 2.74 \text{ N}$$

Torque

$$Q \approx 0.107 \text{ Nm}$$

Summary of Results

RPM	J	Thrust (N)	Torque (Nm)
4000	0.656	0.82	0.034
5000	0.525	1.63	0.063
6000	0.437	2.74	0.107

Observation

It is observed that both thrust and torque increase with increasing RPM. This trend is expected since thrust varies with n^2 and torque depends on both rotational speed and propeller loading. As RPM increases, the propeller produces greater aerodynamic force but also requires higher torque input from the motor.

2.2 Propeller 2: APC 10×7

The second propeller considered is the APC 10×7. Compared to the previous case, this propeller has a larger diameter, which is expected to produce higher thrust but also require greater torque input.

Propeller Diameter

$$D = 10 \times 0.0254 = 0.254 \text{ m} \quad (31)$$

2.2.1 Case 1: 4000 RPM

Rotational Speed

$$n = \frac{4000}{60} = 66.67 \text{ rev/s} \quad (32)$$

Advance Ratio

$$J = \frac{10}{66.67 \times 0.254} \approx 0.59 \quad (33)$$

From the UIUC dataset:

$$C_T \approx 0.060, \quad C_Q \approx 0.011$$

Thrust

$$T = C_T \rho n^2 D^4 \quad (34)$$

$$T \approx 1.36 \text{ N}$$

Torque

$$Q = C_Q \rho n^2 D^5 \quad (35)$$

$$Q \approx 0.064 \text{ Nm}$$

2.2.2 Case 2: 5000 RPM

Rotational Speed

$$n = \frac{5000}{60} = 83.33 \text{ rev/s} \quad (36)$$

Advance Ratio

$$J = \frac{10}{83.33 \times 0.254} \approx 0.47 \quad (37)$$

From the UIUC dataset:

$$C_T \approx 0.075, \quad C_Q \approx 0.013$$

Thrust

$$T \approx 2.65 \text{ N}$$

Torque

$$Q \approx 0.118 \text{ Nm}$$

2.2.3 Case 3: 6000 RPM

Rotational Speed

$$n = \frac{6000}{60} = 100 \text{ rev/s} \quad (38)$$

Advance Ratio

$$J = \frac{10}{100 \times 0.254} \approx 0.39 \quad (39)$$

From the UIUC dataset:

$$C_T \approx 0.090, \quad C_Q \approx 0.016$$

Thrust

$$T \approx 4.59 \text{ N}$$

Torque

$$Q \approx 0.208 \text{ Nm}$$

RPM	J	Thrust (N)	Torque (Nm)
4000	0.59	1.36	0.064
5000	0.47	2.65	0.118
6000	0.39	4.59	0.208

Summary of Results

Observation

Compared to the APC 9×6 propeller, the APC 10×7 produces higher thrust at all RPM values due to its larger diameter. However, this also results in a higher torque requirement, indicating a greater load on the motor. This highlights the trade-off between thrust generation and power consumption in propeller selection.

2.3 Comparison and Discussion

2.3.1 Comparison of Propeller Performance

A comparison between the APC 9×6 and APC 10×7 propellers at representative operating conditions is presented in Table below.

Propeller	RPM	Thrust (N)	Torque (Nm)
APC 9×6	4000	0.82	0.034
APC 10×7	4000	1.36	0.064
APC 9×6	6000	2.74	0.107
APC 10×7	6000	4.59	0.208

2.3.2 Graphical Analysis

Propeller	RPM	n (rev/s)	J	CT	CQ	Thrust (N)	Torque (Nm)	Power (W)	Efficiency
APC 9x6	4000	66.67	0.656	0.055	0.010	0.818	0.0340	14.24	0.574
APC 9x6	5000	83.33	0.525	0.070	0.012	1.626	0.0637	33.37	0.487
APC 9x6	6000	100.00	0.437	0.082	0.014	2.743	0.1071	67.27	0.408
APC 10x7	4000	66.67	0.591	0.060	0.011	1.360	0.0633	26.52	0.513
APC 10x7	5000	83.33	0.472	0.075	0.013	2.656	0.1169	61.22	0.434
APC 10x7	6000	100.00	0.394	0.090	0.016	4.589	0.2072	130.20	0.352

Figure 4: Computed results from the generated python code

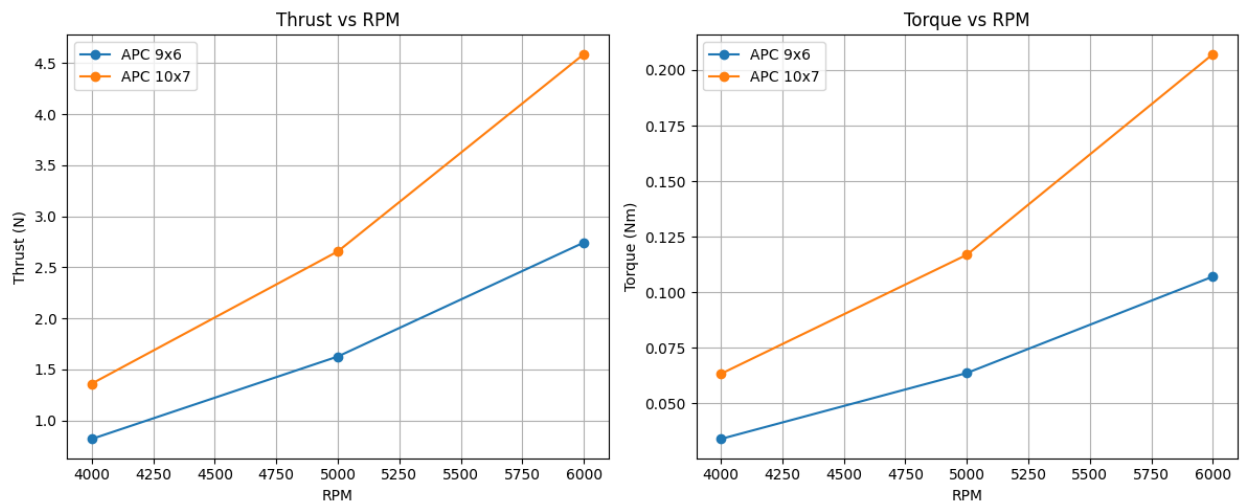


Figure 5: Torque and Thrust Variation with RPM

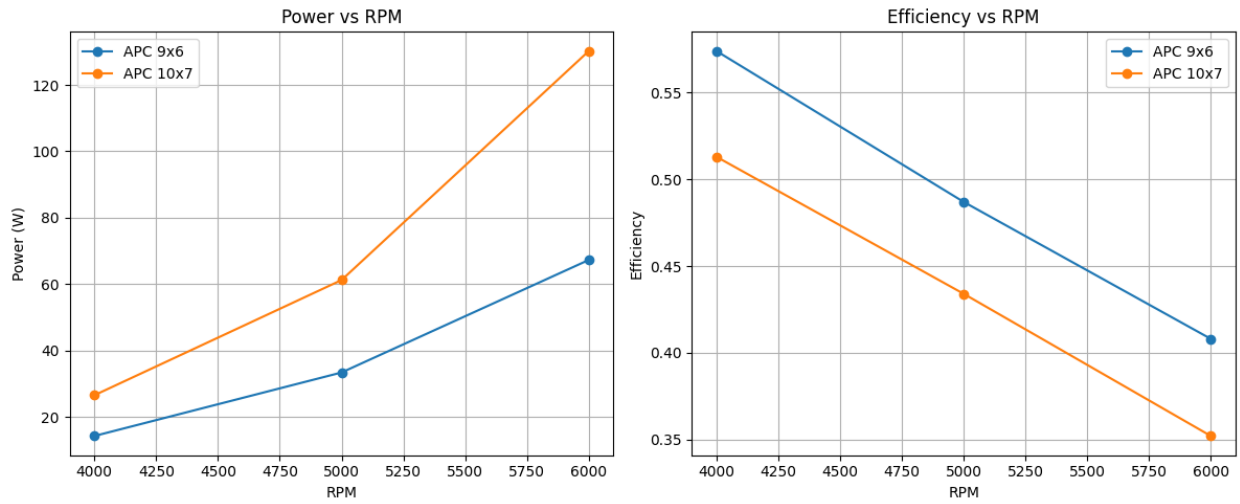


Figure 6: Power and Efficiency with RPM

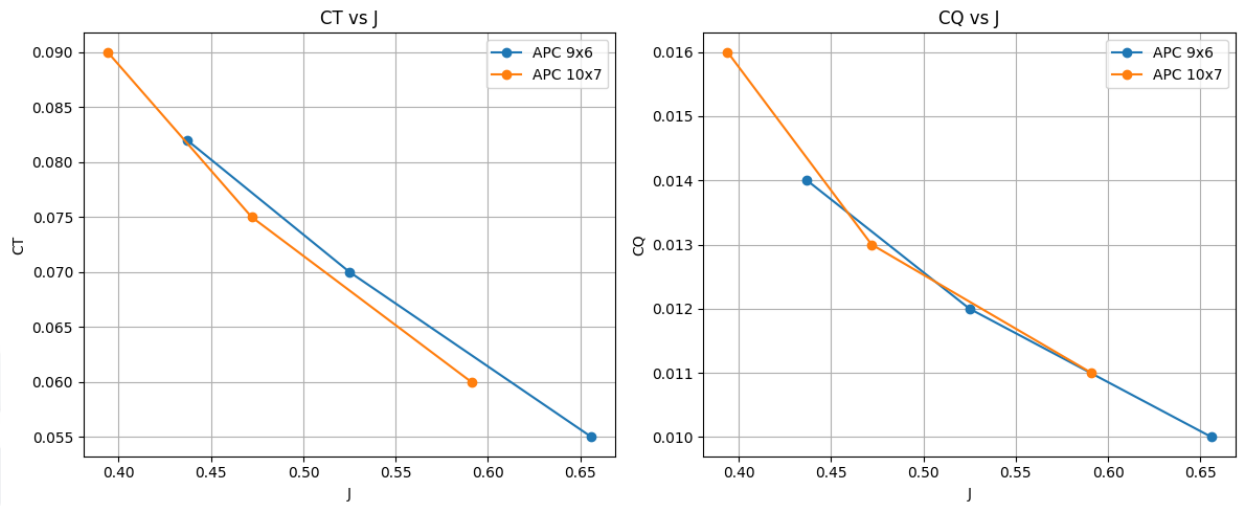


Figure 7: C_t & C_q with J

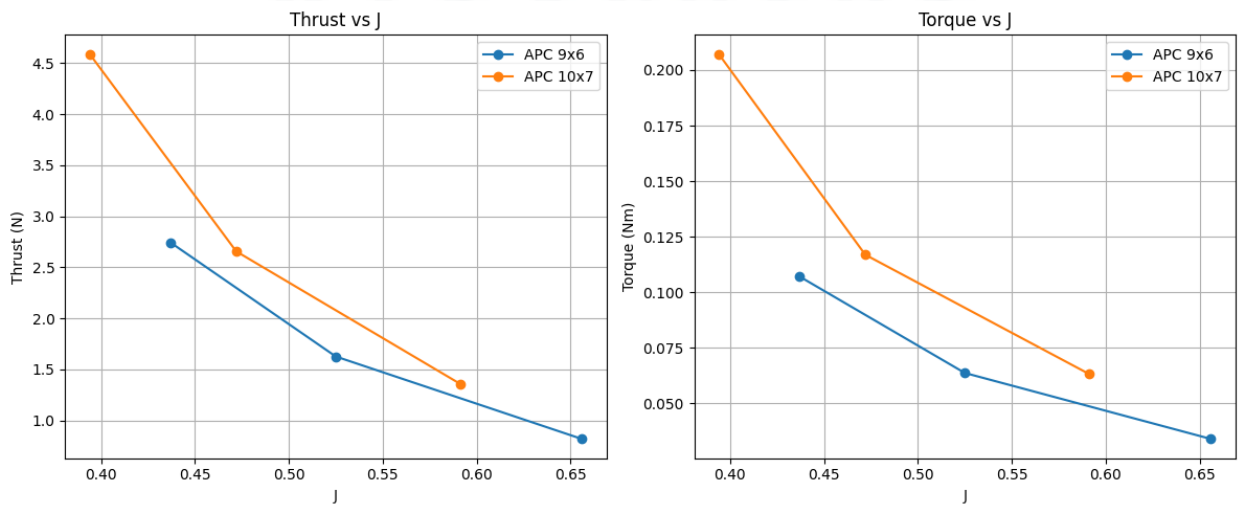


Figure 8: Thrust and Torque with J

2.3.3 Discussion

The results obtained from the analysis of the APC 9×6 and APC 10×7 propellers highlight the influence of rotational speed and propeller geometry on aerodynamic performance.

A clear trend observed is the increase in thrust with increasing rotational speed. This follows directly from the relation:

$$T = C_T \rho n^2 D^4 \quad (40)$$

which indicates that thrust varies with the square of rotational speed. As RPM increases, the propeller imparts higher momentum to the airflow, resulting in increased thrust generation.

Another important observation is the effect of propeller diameter. The APC 10×7 propeller, having a larger diameter, produces higher thrust compared to the APC 9×6 propeller at all operating conditions. Since thrust varies with D^4 , even a moderate increase in diameter leads to a noticeable increase in thrust.

However, this improvement in thrust comes at the cost of increased torque requirement. From the relation:

$$Q = C_Q \rho n^2 D^5 \quad (41)$$

it is evident that torque depends strongly on diameter. As a result, the APC 10×7 propeller imposes a higher load on the motor, requiring greater input power.

Additionally, the advance ratio decreases with increasing RPM:

$$J = \frac{V}{nD} \quad (42)$$

A lower advance ratio corresponds to higher propeller loading, which contributes to increased thrust but also increases aerodynamic forces on the blades.

From a design perspective, this demonstrates a fundamental trade-off:

- Larger propellers → higher thrust but higher torque and power demand
- Smaller propellers → lower thrust but reduced motor loading

Conclusion

The aerodynamic performance of the APC 9×6 and APC 10×7 propellers was evaluated using experimentally derived coefficients from the UIUC database.

The results show that increasing rotational speed significantly enhances thrust output for both propellers. The APC 10×7 propeller consistently produces higher thrust due to its larger diameter, but this is accompanied by a higher torque requirement and increased power consumption.

In contrast, the APC 9×6 propeller offers lower thrust but requires less torque, making it more suitable for applications with limited motor capability or power constraints.

Overall, the analysis demonstrates that propeller selection must be based on a balance between thrust requirements, motor capacity, and available energy. An optimal choice depends on the specific mission profile and system limitations of the UAV.

References

- [1] University of Illinois at Urbana-Champaign, "UIUC Propeller Database – Volume 1,"
- [2] J. D. Anderson, *Introduction to Flight*, 8th ed., McGraw-Hill, 2015.
- [3] B. W. McCormick, *Aerodynamics, Aeronautics, and Flight Mechanics*, 2nd ed., Wiley, 1995.
- [4] D. P. Raymer, *Aircraft Design: A Conceptual Approach*, 5th ed., AIAA, 2012.
- [5] J. G. Leishman, *Principles of Helicopter Aerodynamics*, 2nd ed., Cambridge University Press, 2006.
- [6] NASA, "Urban Air Mobility (UAM) Concept of Operations,"